BTS-VIII(S)–08.18–1135 Reg. No.

C

B.Tech. Degree VIII Semester Supplementary Examination August 2018

ME 802 COMPRESSIBLE FLUID FLOW

(2006 Scheme)

Time: 3 Hours Maximum Marks: 100

(Approved gas tables are permitted)

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Derive the steady flow energy equation.

(b) What is meant by geometric choking? Explain.

(c) What is Mach wave? Explain its characteristics.

(d) Derive the Prandtl's relation for oblique shock wave.

(e) What is meant by friction choking? Explain.

(f) Derive an expression for the maximum possible heat transfer (Q*) in a Rayleigh flow.

(g) Derive the Rayleigh supersonic pitot formula.

(h) Explain the working principle of Prandtl Pitot static tube with the help of a neat figure.

PART B

(4 × 15 = 60)

II. (a) Define Mach angle and Mach cone. (5)

(b) A jet plane is flying at 10 km with a cabin pressure of 101 kPa and a cabin temperature of 20°C. Suddenly, a bullet is fired inside the cabin and pierces the fuselage; the resultant hole is 2 cm in diameter. Assume that the temperature within the cabin remains constant and that the flow through the hole behaves as that through a converging nozzle with an exit diameter of 2 cm. Take the cabin volume to be 100 m^3. Calculate the time for the cabin pressure to decrease to half the initial value. At 10 km, pressure $P = 26.5$ kPa and temperature $T = 223.3$ K. (10)

OR

III. (a) Derive the area-velocity relation for a compressible flow. (5)

(b) A converging-diverging nozzle is designed to operate isentropically with an exit Mach number of 1.5. The nozzle is supplied from an air reservoir in which the pressure is 500 kPa and temperature is 400 K. The nozzle throat area is 5 cm^2. Assume air to behave as a perfect gas, with $\gamma = 1.4$ and $R = 287$ J/kgK. Determine the ratio of exit area to throat area. (10)

(P.T.O.)

IV. (a) Derive an expression for the temperature ratio across a normal shock. (5)

(b) A normal shock occurs in an air flow at a point where the velocity is 750 m/s, the pressure is 50 kPa, and the temperature is 10°C. Find the velocity, pressure and static temperature downstream of the shock wave. (10)

OR

V. (a) Derive an expression for the pressure ratio across an oblique shock wave. (5)

(b) Air flowing at a Mach number of 2.5 passes over a wedge that turns the flow through an angle of 5°. Find the pressure ratio across the oblique shock that is generated. (10)

VI. (a) Derive an expression for density ratio along a Fanno flow. (5)

(b) A constant area duct that is 20 cm in length by 2 cm in diameter is connected to a reservoir through a converging nozzle. For a reservoir pressure and temperature of 1 MPa and 500 K, determine the maximum air flow rate in kg/s through the system. Assume f is equal to 0.008 and $\gamma = 1.4$. (10)

OR

VII. (a) Derive an expression for change in entropy along a Rayleigh flow. (5)

(b) Air enters a constant diameter duct at a pressure of 200 kPa. At the exit of the duct the pressure is 120 kPa, the Mach number is 0.75, and the stagnation temperature is 603 K. Determine the inlet Mach number and the heat transfer per unit mass of air. (10)

VIII. (a) Describe the starting and steady operation of a supersonic wind tunnel. (5)

(b) A supersonic helium wind tunnel is designed with a convergent divergent nozzle. Test section specifications are as follows: (10)

Diameter = 10 cm   Mach number = 3

Static pressure at 15 km altitude = 12.1 kPa

Static temperature at 15 km altitude = 216.7 K

Determine the mass flow rate, and the reservoir temperature and pressure to be maintained. Assume isentropic flow in the nozzle at design conditions. Assume that helium behaves as a perfect gas with $\gamma = 5/3$ and $R = 2.077$ kJ/kgK.

OR

IX. (a) A pitot tube is placed in a stream of carbon dioxide in which the pressure is 60 kPa and the Mach number is 3. What will the pitot pressure be? (6)

(b) Explain the working of the following with the help of neat figures. (9)

(i) Interferometer   (ii) Hot wire anemometer

(iii) Schlieren apparatus.

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